Time to Diagnosis: Predictive Modeling for Lung Cancer Screening

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Background and Motivation

Challenges about Lung Cancer

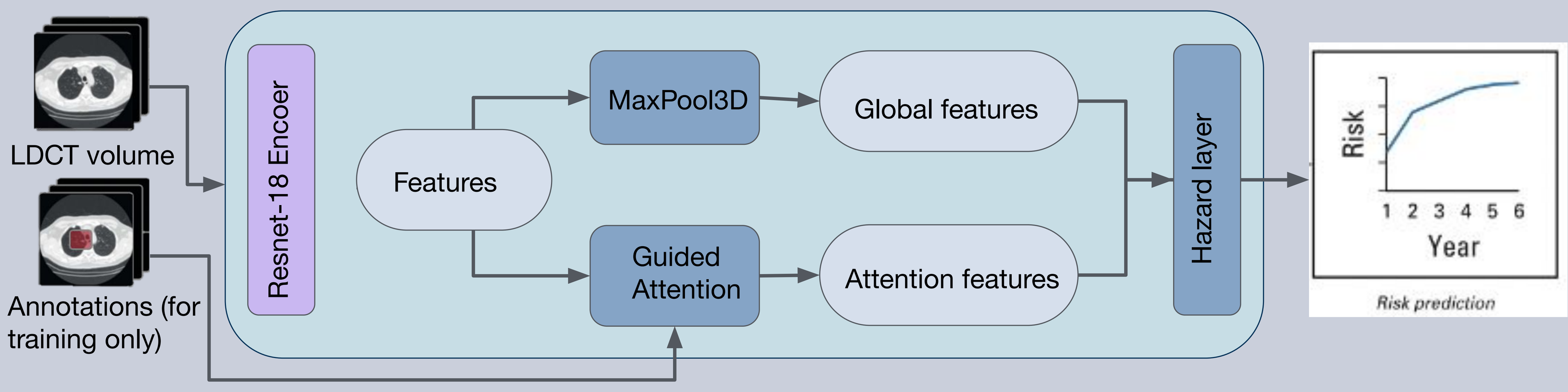
- Leading cause of cancer-related mortality[1]
- Symptoms can appear late, early diagnosis is important but difficult

Necessity for a Deep Learning solution:

- Screening Intervals: Minimizing intervals reduces radiation and FP
- Frequent CT screenings risk overdiagnosis and require biopsy confirmation
- **Individualized Assessment**: Tailor risk assessments and screening recommendations using deep learning cancer risk models

Sybil as baseline model[2]

- Predict lung cancer risk up to 6 years using a single LDCT scan
- Pretrained on Kinetics-400 dataset
- **Generalization Limitations:** Pretrained on a general video dataset (Kinetics-400), Sybil may not be as finely tuned for medical imaging



Overall goal: Use ResNet-18 3D as the baseline encoder in Sybil, experiment with adapting 2D pretrained models to 3D using ACS Convolution, and assess the impact of domain-specific pretraining on model performance with lung CT scans.

Exploring the potential of pretrained 2D encoders transformed to 3D: Leveraging established 2D architectures

a) ResNet-18 Native 3D – Kinetics-400 (baseline encoder)

- Designed for capturing temporal and volumetric information

b) ResNet-18 2D* – ImageNet-1K

* Encoder transformed into 3D using ACS convolutions: 3D convolutions to process slices of CT scans as continuous volume[5]

Evaluating the impact of domain-specific pretrained encoders on embedding representations and performance

Could lead to better initialization, faster convergence, and enhanced embeddings for tasks like predicting future cancer

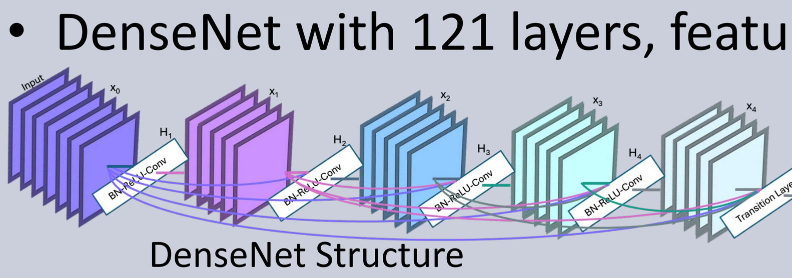
c) ResNet-50 2D* – ImageNet-1K

d) ResNet-50 2D* – RadImageNet

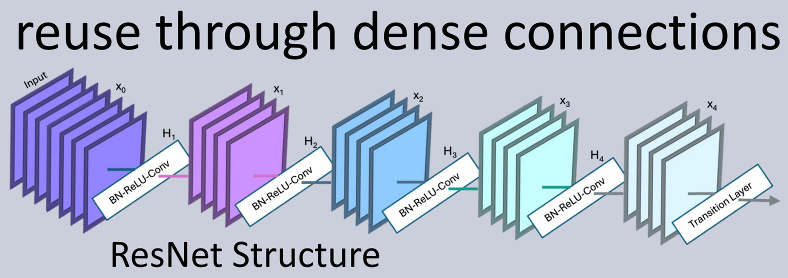
e) DenseNet-121 2D* – ImageNet-1K

f) DenseNet-121 2D* – RadImageNet

- DenseNet with 121 layers, feature reuse through dense connections



DenseNet Structure
$a^{[l]}=g([a^{[0]}, a^{[1]},..., a^{[l-1]}])$



ResNet Structure
$a^{[l]}=g(z^{[l+1]}+a^{[l]})$

Datasets

ImageNet-1K	1.28 million natural images across 1,000 object categories
RadImageNet[5]	1.35 million annotated images of three imaging modalities and covers 14 anatomical areas, including CT lung cancer scans
Kinetics-400	400+ 10-second video clips labeled across 400 human action classes

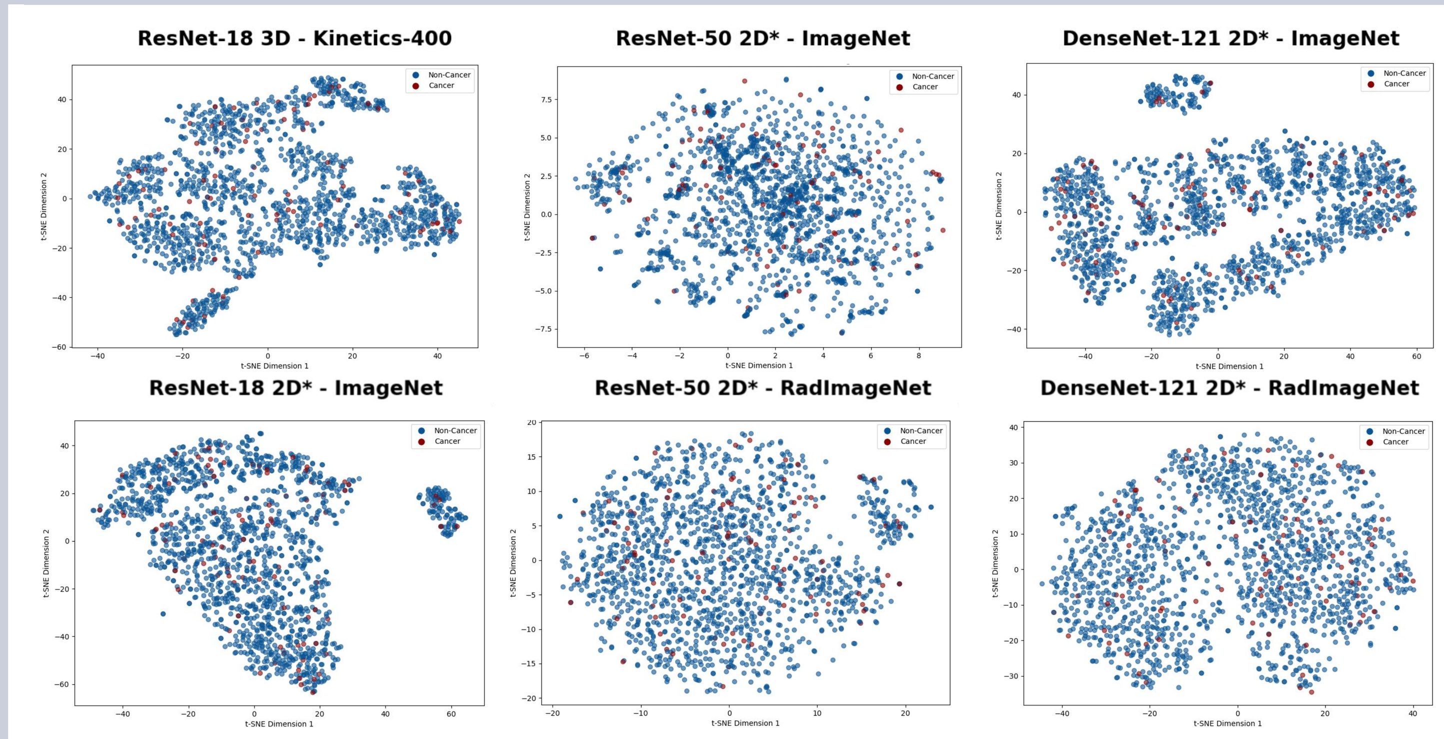
National Lung Screening Trial (NLST) dataset[3]

- Evaluate the efficiency of LDCT in reducing lung cancer mortality
- 53,454 high-risk individuals (current/former smokers aged 55-74)
- Randomized controlled trial comparing LDCT to chest X-ray over 3 years, with a follow-up phase to monitor cancer events (T7)

Preprocessing of subset:

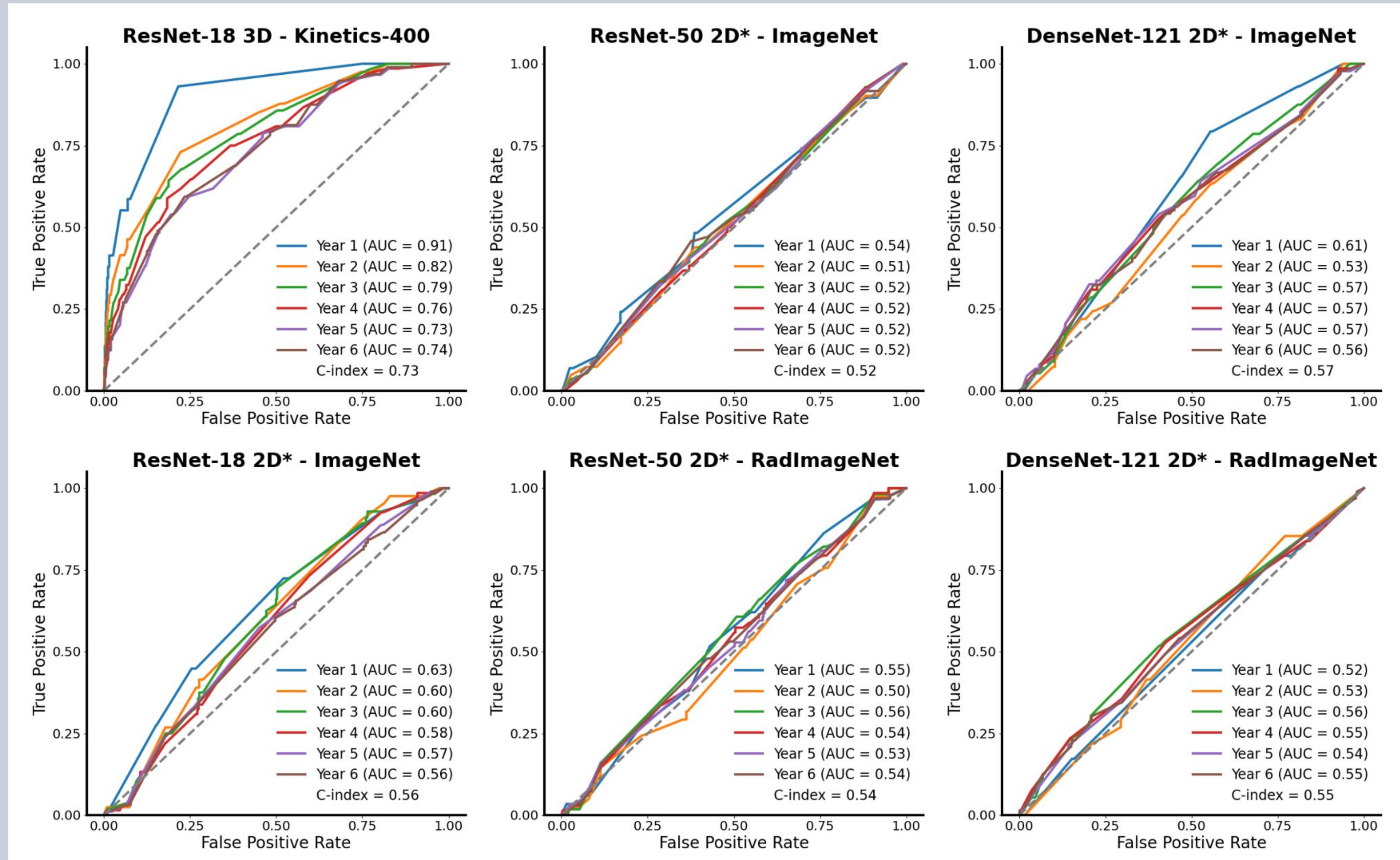
- 25% of NLST data (Train: 6987, Val: 1806, Test: 1561)
- Cancer to non-cancer ratio: (5.3%, 5.65%, 4.74%)
- Converted DICOM images to PNG using OncoData[4]
- Extracted DICOM metadata and nodule annotations across slices

t-SNE embedding space visualization



- **t-SNE Visualization:** t-SNE effectively highlights distinct clustering patterns among various encoders, illustrating how each model uniquely captures and represents features from lung CT scans

ROC curves of different encoders



- **ROC curves:** Showcase Sybil's capacity to predict future lung cancer from a single CT scan over a six-year period
- **C-index:** Predictive accuracy of risk prediction, (ability to correctly rank the order of events)
- **AUC**: Indicates the overall performance of a model; a higher AUC signifies better discriminative ability

Main Findings

- **2D-to-3D models:** ResNet-18 3D consistently outperforms 2D-to-3D models, likely due to better-optimized hyperparameters and training protocols, which are less effective for other architectures
- **General vs. Domain-Specific Pretraining:** RadImagenet models show no significant performance improvement over ImageNet models, indicating limited benefit from domain-specific pretraining
- **t-SNE Visualization:** The lack of clear separation might indicate that the features extracted by the encoders are not discriminative enough to differentiate between cancerous and non-cancerous samples effectively

References

[1] Siegel, Rebecca L., Angela N. Giaquinto, and Ahmedin Jemal. "Cancer statistics, 2024." CA: a cancer journal for clinicians 74.1 (2024): 12-49.
[2] Mikhael, Peter G., et al. "Sybil: a validated deep learning model to predict future lung cancer risk from a single low-dose chest computed tomography." Journal of Clinical Oncology 41.12 (2023): 2191.
[3[National Lung Screening Trial Research Team et al. "The National Lung Screening Trial: overview and study design." Radiology vol. 258,1 (2011): 243-53. doi:10.1148/radiol.10091808.
[4] https://github.com/yala/OncoData_Public/tree/sybil
[5] Yang, Jiancheng, et al. "Reinventing 2d convolutions for 3d images." IEEE Journal of Biomedical and Health Informatics 25.8 (2021): 3009-3018.
[6] Mei, Xueyan, et al. "RadImageNet: an open radiologic deep learning research dataset for effective transfer learning." Radiology: Artificial Intelligence 4.5 (2022): e210315.

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